LXR
Owners Manual

Fork v1.0 (based on LXR firmware v0.91)

This manual covers hardware overview, sequencer operation, voice synthesis parameters, MIDI, and firmware updates.

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1 Overview

In this first chapter we will focus on the physical appearance of the LXR, describing the front panel controls as well as the connection jacks on the back. The basic menu navigation is also explained.

1.1 The Front

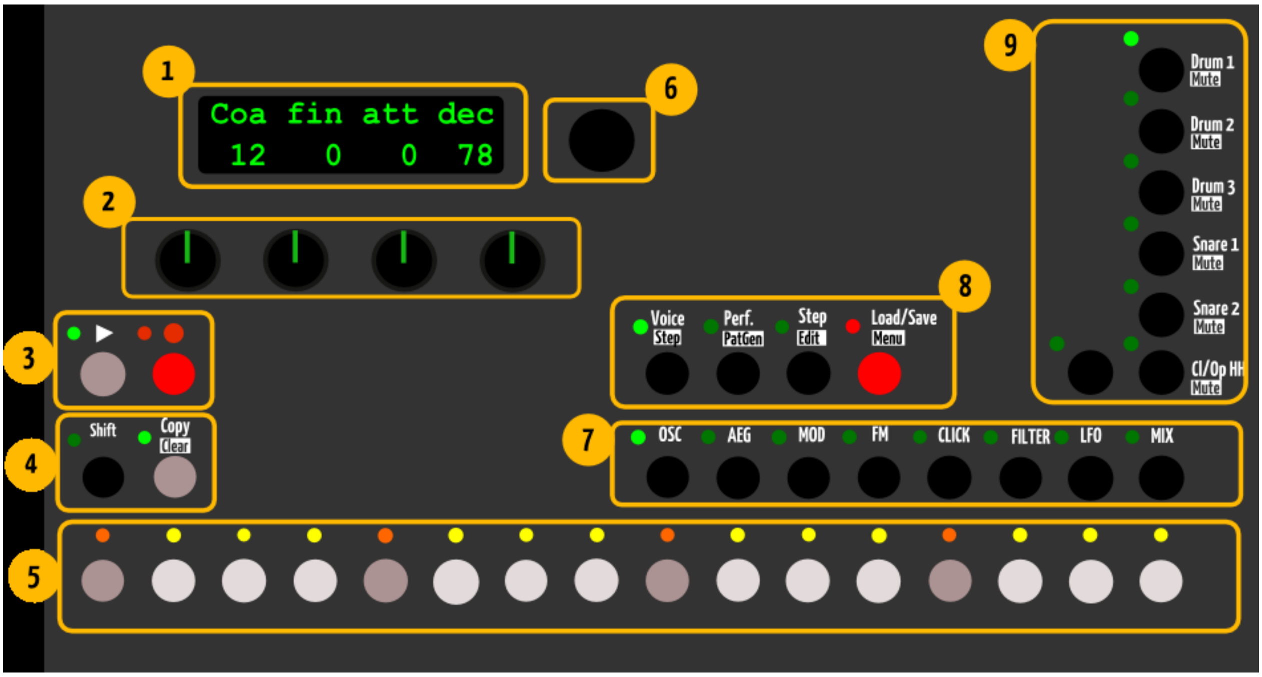

Figure 1. LXR front panel with numbered control regions.

1.1.1 1) Display

The display shows parameter values from the selected menu page.

1.1.2 2) Knobs

The 4 knobs edit the values shown in the display. A parameter fetch option — which prevents value jumps when switching menu pages — can be activated in the settings menu.

1.1.3 3) Record/Play Buttons

The Play button starts and stops playback. The current pattern is reset to the first step when the sequencer is stopped.

The Record button activates the recording function. When the LED is lit, all incoming MIDI notes are recorded into the active pattern. Knob movements are also recorded to the selected automation track.

1.1.4 4) Shift Button

Activates an alternate function set for the buttons.

1.1.5 4) Copy/Clear Button

Used to copy and clear sequencer data.

1.1.6 5) 16 Sequencer Buttons

Function depends on the active mode:

- **Voice mode:** set/reset the 16 main steps in the sequencer.
- **Performance mode:** buttons 1–7 trigger manual rolls. Buttons 8–16 are unused.
- **Step Edit mode:** select one of the 16 main steps for editing.
- **Menu mode:** no function.

1.1.7 6) Encoder

Used to navigate through the menu.

1.1.8 7) 8 Select Buttons

Function depends on the active mode:

- **Voice Edit mode:** select the synth section to edit (Oscillators, Mix, Envelopes, etc.)
- **Performance mode:** change the playing pattern.
- **Step Edit mode:** select/edit sub steps.
- **Menu/Pattern generator mode:** no function.

1.1.9 8) 4 Mode Buttons

Switch between the 4 operating modes. Modes shown in brackets are selected using the shift button.

- Button 1 — Voice edit mode
- Button 2 — Performance mode [Pattern generator mode]
- Button 3 — Step Edit mode
- Button 4 — Load/Save [Settings menu]

1.1.10 9) 7 Voice Buttons

The voice buttons select the active track to edit. Together with the shift key they mute/unmute the voices. They behave the same in all operation modes except performance mode — see the performance mode chapter for details.

1.2 The Back

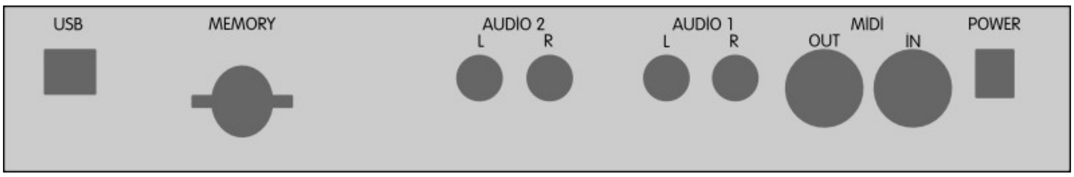

Figure 2. LXR rear panel connections.

1.2.1 USB

The synthesizer provides a class-compliant USB MIDI interface.

1.2.2 Memory

Insert an SD card to load and save preset and pattern data. The card must be inserted with the label facing downward.

> **Attention!** The SD card must use a FAT32 filesystem for reliable operation. The synth will work with FAT16 cards too, but **firmware updates are only possible from FAT32 cards**.

1.2.3 4 Audio Outputs

4 mono audio outputs at line level. Can also be used as 2 stereo output pairs.

1.2.4 MIDI

MIDI in and output via standard DIN connectors.

1.2.5 Power Connector

7–9V DC power plug, **centre pin positive, at least 600 mA**.

1.3 Menu Navigation

The upper row of the display shows the short name of the parameter. The bottom row shows its current value. Use one of the 4 knobs to change it.

```
frq   Res   typ   drv >
32     120   LP     0
```

- The encoder selects a menu parameter. A capital letter at the beginning of the name indicates the selected parameter. In the example above, `Res` is selected.
- If there are more than 4 entries in the current menu, a `>` or `<` sign in the upper-right corner indicates a second page. The two pages toggle by pressing the menu button again.

Pushing the encoder opens the detail page:

```
Filter
Resonance                    120
```

On the detail page you can see the full parameter name and change its value with the encoder. This is useful for fine adjustments where the knob control is too coarse. Push the encoder again to return to the normal menu mode.

2 Quickstart

The fastest way to get beats out of your new box:

1. Download the factory SD card image from: `http://sonic-potions.com/public/SdCardImage.zip`
2. Unzip all files to the root folder of your (FAT32!) SD card.
3. Insert the card and power up the synth.
4. Press the Play button. The chaselight starts running.
5. Use the 16 step buttons to activate some steps. You should now hear the first voice playing.
6. Use the 7 voice select buttons to change the active track/voice.
7. Again, use the 16 step buttons to activate some steps. The next voice should now play as well.
8. Press the Load/Save mode button once. You are now in preset load mode.
9. Use the encoder to select another preset. The sound changes as soon as you move the encoder.

3 Voices

The LXR offers 6 voices, each optimised for different drum sounds. The type of a voice cannot be changed. There are 3 drum voices, a subtractive clap/snare voice, an FM percussion voice, and a hi-hat voice.

3.1 Drum Voices (voices 1–3)

The drum voices are suited to — but not limited to — kick drums, toms, cowbells, and other percussive sounds.

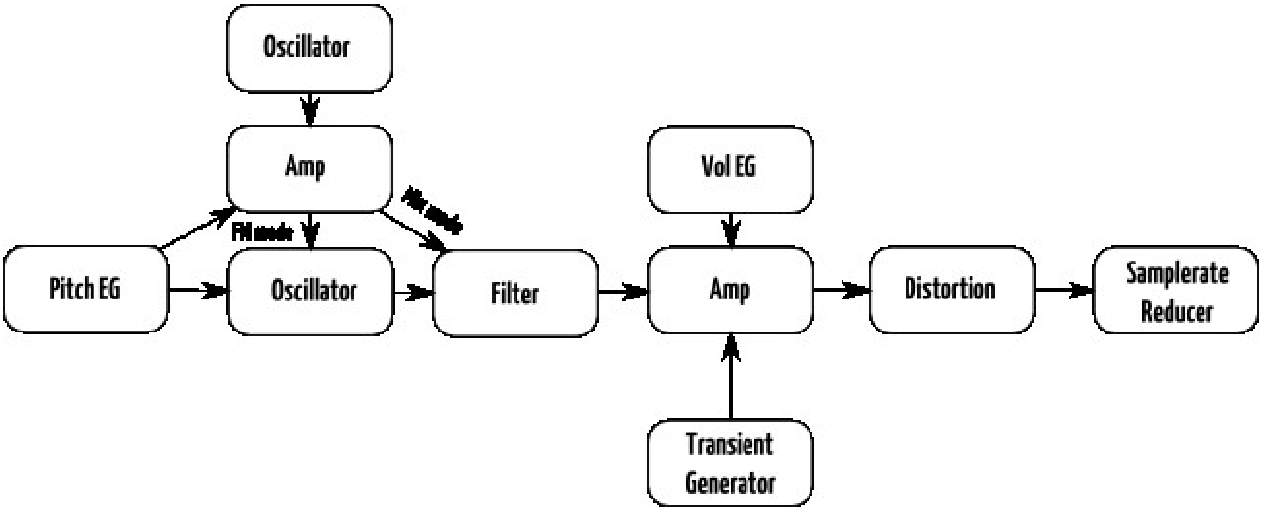

Figure 3. Drum voice signal flow (voices 1–3).

3.2 Snare Voice (voice 4)

This voice is good for snare drum and clap sounds. A noise source and a pitched oscillator can be mixed. Only the noise source is routed through the filter. There is no FM capability on this voice.

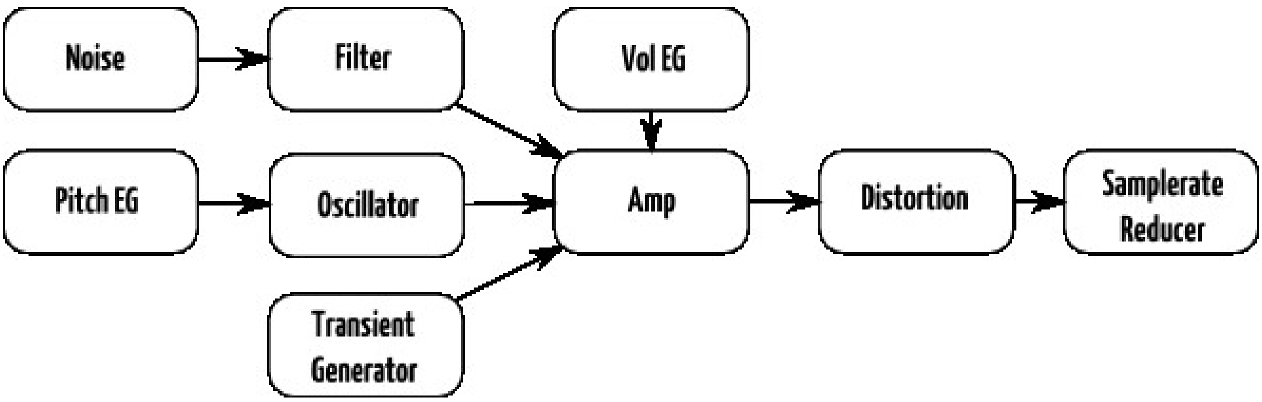

Figure 4. Snare voice signal flow (voice 4).

3.3 Cymbal/Clap Voice (voice 5)

This voice uses a 3-operator FM synthesis to generate the sound. The main oscillator is modulated by 2 modulation oscillators. It is excellent for metallic and noisy sounds.

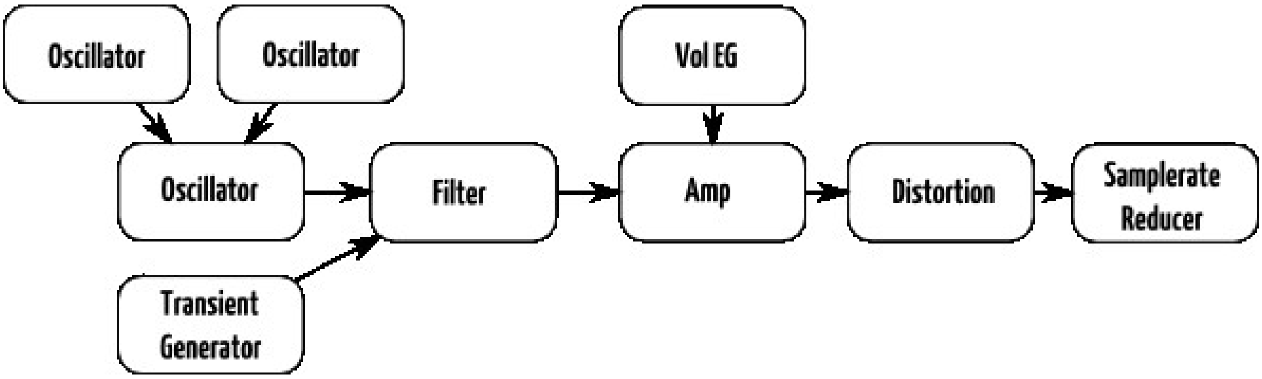

Figure 5. Cymbal/Clap voice signal flow (voice 5).

3.4 Hihat Voice (voice 6)

The hihat voice is nearly identical to the Cymbal voice, but it offers 2 different decay times for the amplitude envelope. The voice is shared between sequencer tracks 6 and 7, each using one of the 2 available decay times. This allows open and closed hihats to be played. The closed hat (track 6) chokes the open hat (track 7).

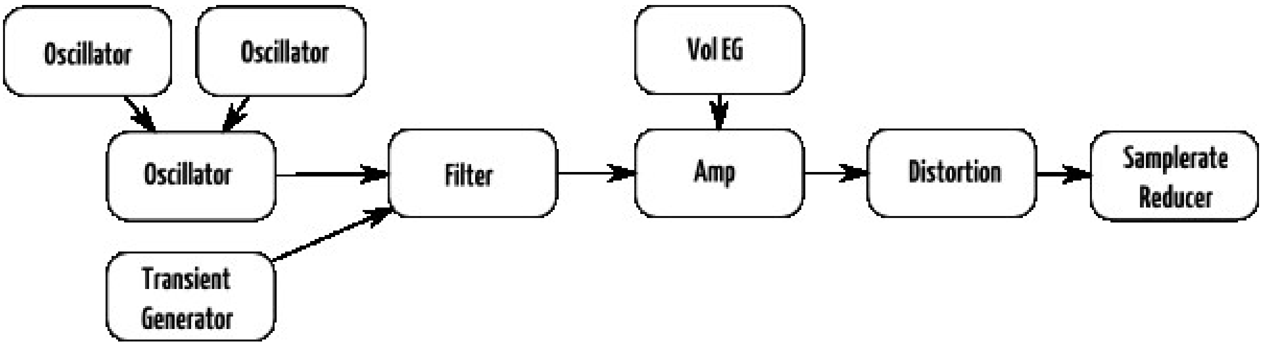

Figure 6. Hihat voice signal flow (voice 6).

3.5 Voice Parameter Menu Sections

The synthesis parameters of each voice can be altered in voice edit mode. The synthesis engine is grouped into 8 sections selectable with the select buttons.

3.5.1 Oscillator Page (OSC)

Provides access to the main oscillator parameters — frequency and waveform.

Drum 1–3, Clap, Hihat

Display	Name	Description
`coa`	Coarse tune	Coarse tuning of the main oscillator in semitones.
`fin`	Fine tune	Fine tuning of the main oscillator. ±50 cents.
`wav`	Waveform	The waveform of the main oscillator.

Snare

Display	Name	Description
coa	Coarse tune	Coarse tuning of the main oscillator in semitones.
fin	Fine tune	Fine tuning of the main oscillator. ±50 cents.
noi	Noise frequency	Coarse tuning of the noise oscillator.
mix	Oscillator mix	Mix ratio between oscillator and noise source.
wav	Waveform	The waveform of the main oscillator.

3.5.2 Amplitude Envelope Page (AEG)

Each voice has an amplitude envelope. The common parameters are attack time, decay time, and slope.

Drum 1–3

Display	Name	Description
att	Amplitude envelope attack	Rise time of the envelope.
dec	Amplitude envelope decay	Fall time of the envelope.
slp	Amplitude envelope slope	Variable slope from exponential to linear to logarithmic.

Snare/Cymbal

Display	Name	Description
att	Amplitude envelope attack	Rise time of the envelope.
dec	Amplitude envelope decay	Fall time of the envelope.
rpt	Repeat count	Number of retriggers.
slp	Amplitude envelope slope	Variable slope from exponential to linear to logarithmic.

HiHat

Display	Name	Description
att	Amplitude envelope attack	Rise time of the envelope.
d1	Closed hihat decay time	Fall time of the envelope for the closed hihat.
d2	Open hihat decay time	Fall time of the envelope for the open hihat.
slp	Amplitude envelope slope	Variable slope from exponential to linear to logarithmic.

3.5.3 Modulation Page (MOD)

Contains velocity modulation parameters and, where available, the second envelope parameters.

Drum 1–3, Snare

Display	Name	Description
dec	Modulation envelope decay	Fall time of the modulation envelope.
slp	Modulation envelope slope	Variable slope from exponential to linear to logarithmic.
mod	Modulation envelope amount	Controls how strongly the envelope modulates its target. The envelope is hardwired to pitch; this controls the modulation intensity. In FM mode the EG is additionally hardwired to the FM amount.
dst	Velocity mod. destination	The note velocity can be routed to any voice parameter. Select the target from a list.
amt	Velocity mod. amount	Amount of the velocity modulation.
vol	Velocity to volume	On or Off. When active, note velocity controls voice volume.

Clap, Hihat

Display	Name	Description
dst	Velocity mod. destination	Route note velocity to any voice parameter.
amt	Velocity mod. amount	Amount of the velocity modulation.
vol	Velocity to volume	On or Off. When active, note velocity controls voice volume.

3.5.4 Frequency Modulation Page (FM)

Hosts frequency, waveform, and modulation amount settings for the FM oscillators.

Drum 1–3

Display	Name	Description
amt	Modulation amount / mix ratio	FM mode: modulation amount. Mix mode: mix ratio of the 2 oscillators.
frq	Frequency of FM OSC	Coarse tuning of the FM oscillator in semitones.
wav	Waveform of FM OSC	Waveform of the FM oscillator.
mod	Oscillator mode	FM mode: main OSC is modulated by FM OSC. Mix mode: main OSC and FM OSC are mixed.

Snare — Voice 4 has no FM capability; this page is empty.

Clap, Hihat

Display	Name	Description
f1	Frequency 1	Frequency of the first modulation oscillator.
f2	Frequency 2	Frequency of the second modulation oscillator.
g1	Gain 1	Gain of the first modulation oscillator.
g2	Gain 2	Gain of the second modulation oscillator.
wav	Waveform 1	Waveform of the first modulation oscillator.
wav	Waveform 2	Waveform of the second modulation oscillator.

3.5.5 Transient Generator Page (Click)

Parameters of the transient generator: waveform, volume, and frequency of a short attack sample mixed into the voice output.

Display	Name	Description
wav	Transient wave shape	Selects the transient sample to play.
vol	Transient volume	Volume of the transient sample.
frq	Transient frequency	Frequency of the transient sample.

3.5.6 Filter Page (FIL)

All filter parameters on one page.

Display	Name	Description
frq	Filter frequency	Changes the cutoff frequency of the filter.
res	Filter resonance	Adjusts the filter resonance.
typ	Filter type	Selects the filter characteristic.
drv	Filter drive	Increases the input gain of the filter.

3.5.7 LFO Page (LFO)

Each voice has an LFO editable on this page.

Display	Name	Description
`frq`	LFO frequency	Manual LFO rate control. Only available when sync is off.
`snc`	LFO sync	Activates clock sync. Sync rates: $\frac{1}{2}$, $\frac{1}{4}$, etc.
`mod`	Modulation amount	Controls how strongly the LFO signal affects the target.
`wav`	LFO waveform	Sine (sin), Triangle (tri), Saw up (sup), Saw down (sdn), Square (sqr), Random (rnd), Exponential saw up (xup), Exponential saw down (xdn).
`rtg`	LFO retrigger	Allows the LFO to be retriggered from a sequencer track (v1–v6). The LFO resets its phase whenever a note plays on the selected track.
`off`	Phase offset	Phase offset applied when the LFO is retriggered.
`voi`	Destination voice	Select the target voice for LFO modulation.
`dst`	Destination select	Select the parameter to modulate (list of all parameters for the selected target voice).

3.5.8 Mixer Page (Mix)

Volume, panning, routing, voice effects, and sequencer track length.

Display	Name	Description
`vol`	Voice volume	Adjusts the volume of the voice.
`pan`	Voice panning	Voice panning. Only active when output is set to a stereo channel.
`sr`	Sample rate	Sample rate decimation.
`drv`	Drive	Soft clipping amount.
`out`	Output	Selects the hardware audio output for this voice: one of the 2 stereo channels, or one of the 4 individual mono outputs.
`len`	Track length	The length of the sequencer track in $\frac{1}{16}$ steps.

4 Sequencer

The LXR has a sequencer for playing and programming drum patterns. The basic functionality is similar to classic TR-x0x machines, with additional features.

4.1 Tracks

There are 7 sequencer tracks — one per voice, plus an additional track to control the open hihat of voice 6.

4.1.1 Select Tracks

The active track is changed with the 7 voice buttons. A lit LED next to the button shows the selected track.

4.1.2 Mute Tracks

Hold the shift button and press a voice button to mute/unmute that track. The LEDs of unmuted voices light up.

> **Did you know?** In performance mode the function of the voice buttons is inverted — they mute/unmute without needing to hold shift. This allows single-handed operation.

4.1.3 Unmute All Tracks

To unmute all muted tracks simultaneously, go to performance mode and hold shift. The voice buttons then unmute all tracks up to and including the pressed button.

Example:

- Shift + Voice Button 4 = tracks 1, 2, 3 and 4 will be unmuted.
- Shift + Voice Button 7 = all tracks are unmuted.

4.2 Steps

There are 16 steps per track. Steps are set or removed using the 16 sequencer buttons.

4.2.1 Sub Steps

Each of the 16 main steps consists of 8 sub steps, giving a total of 128 steps per pattern.

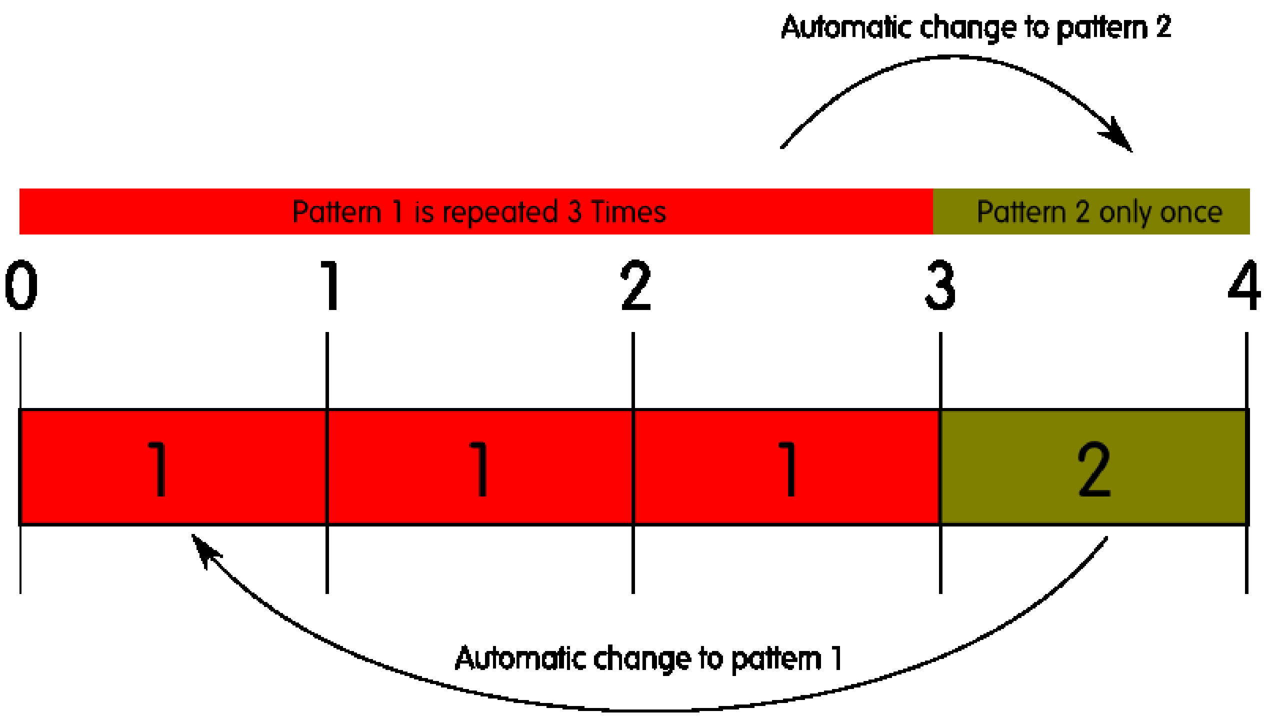

Figure 7. Sub steps shown within the step buttons.

Sub steps define a step pattern played when the corresponding main step is active. They can be used to create flam or roll sounds on a particular step, or to place a note between the $\frac{1}{16}$th notes of the main steps.

If the main step is inactive, none of its sub steps will be played. Each sub step can have its own pitch, volume, probability, and automation settings.

4.3 Pattern

A set of all 7 tracks is called a pattern. Different patterns can be chained to play one after another — see the Performance Mode section for details.

4.4 Pattern Set

A pattern set is a group of 8 patterns saved together on the SD card and directly accessible in performance mode.

4.5 Basic Pattern Editing

Basic pattern editing is done in voice edit mode. Select the track with the voice buttons. The step data is displayed on the 16 sequencer button LEDs. Use the sequencer buttons to set and remove steps.

4.6 Step Parameters

Each step has a set of parameters. To edit them in voice edit mode, hold the shift button. The display then shows the step parameter page and the LED of the selected step starts flashing.

4.6.1 Available Step Parameters

Display	Name	Description
vel	Step Velocity	Note-on velocity (0–127). Can be used as a mod source on the voice modulation page.
nte	Step Note	MIDI note number to play (–63 to +63). Default 0 = MIDI note 63.
prb	Step Probability	Chance of the step being played. 0 = never (0%), 63 = 50%, 127 = always (100%).

4.6.2 Step Probability

The step probability value sets the chance of a step being played, from 0 to 127 (0–100%).

On each main step, the sequencer generates a new random number between 0 and 127. If the random number is less than or equal to the step's probability value, the step plays.

Did you know? The random number is generated once per main step and shared across all 8 of its sub steps. This allows interesting sub-step cluster randomisation — the 1st sub step always plays, the 5th plays with 50% chance, and the 3rd and 7th play with 25% chance.

4.7 Automation

Parameter values can be recorded to the sequencer steps. Each step can store 2 parameter changes. They only affect that single step and are ignored if the step is inactive. The modulation targets can be different on every step.

4.7.1 Choosing the Active Automation Track

There are 2 automation tracks per voice. Select the active track before recording.

- Hold shift and press the record button to open the recording options menu.
- The **trk** parameter selects the active automation track.

4.7.2 How to Assign Parameter Automations

Realtime recording:

1. Press the record button. The record LED lights up.
2. While the pattern is playing, tweak sound parameters with the knobs. Changes are recorded to the played steps.

Attention! Activate the quantisation grid before recording automations. Without quantisation, data is recorded to all 128 sub steps, and most of it will land on inactive sub steps if you are only using the 16 main steps.

Lock parameter values to a single step:

1. In voice edit mode, hold down the step button.
2. The step LED starts blinking.
3. Use one of the 4 knobs to assign a parameter value to that step.
4. Release the step button.

Edit automation manually via the step edit menu:

1. Go to the step parameter menu (voice mode: shift + step button).
2. Use the encoder to scroll to page 2. You will see:

Display	**Name**	**Description**
`p1d`	Parameter 1 Destination	Select the target voice parameter from a list.
`p1v`	Parameter 1 Value	Set the parameter to this value on this step.
`p2d`	Parameter 2 Destination	Select the target voice parameter from a list.
`p2v`	Parameter 2 Value	Set the parameter to this value on this step.

Manual editing is the only way to assign parameter automations from one track to another. Edit the target parameter in detail mode (push encoder) to see full names and destination voice.

4.7.3 Clear Automation Data

- Hold shift and press the copy/clear button.
- Use the encoder to select `Clear [autom.1/2]?`
- Press copy/clear again to confirm, or release shift to abort.

4.8 Recording Options Menu

Hold shift and press the record button to open the recording options menu:

```
Trk   qnt
0       16
```

4.8.1 Automation Track

The **`trk`** parameter selects which of the 2 automation tracks is active. Parameter automations are recorded to the active track.

4.8.2 Quantisation

Selects the quantisation grid. All recorded notes and automation data will be quantised to the selected grid. Quantisation cannot be applied retroactively — it must be on before recording.

Parameter value	Description
off	No quantisation
8	Quantise to 1/8 notes
16	Quantise to 1/16 notes
32	Quantise to 1/32 notes
64	Quantise to 1/64 notes

4.9 Copy Sequencer Data

It is possible to copy sequencer tracks and patterns to another position.

1. Press and hold the Copy button. Its LED starts flashing to indicate copy mode is active.
2. Select the source data by pressing a voice button (track copy) or a pattern button (pattern copy). The source LED starts flashing.
3. Press the target voice/pattern button. The data is copied immediately, overwriting the destination.
4. To abort, release the copy button before selecting a target.

4.10 Clear Sequencer Data

Sequencer data can be cleared separately for track, pattern, or automation data.

1. Press Shift + Copy/Clear.
2. The display shows `Clear [track]?`
3. Use the encoder to select what to delete: `[track]`, `[pattern]`, `[autom.1]`, or `[autom.2]`.
4. Press the clear button again to confirm, or release shift to abort.

5 Operation Modes

The synth provides different operation modes, each optimised for a different task. The active mode changes the function of the user interface controls.

5.1 Voice Edit Mode

The voice edit mode is used to modify sound parameters and to do basic pattern editing.

- **Display (1):** parameters of the selected synthesis page.
- **Knobs (2):** edit the visible parameters.
- **Sequencer buttons (5):** set and remove the 16 main steps of the current track.
- **Select buttons (7):** select the active synthesis page (oscillator, envelope, mixer, etc.)
- **Voice buttons (9):** select the voice and sequencer track to edit; with shift, mute single voices.

5.1.1 Pattern Editing

The 16 sequencer buttons set and remove steps in the selected pattern. Each button is a $\frac{1}{16}$th note from a 4/4 bar. A lit LED above the button indicates the step is set.

5.1.2 Step Parameter Editing

1. Press and hold the shift button.
2. The display shows the step parameter page of the selected step.
3. Use the knobs and encoder to change parameters.
4. A flashing LED indicates the selected step.
5. Select a different step using the 16 sequencer buttons.

5.1.3 Sub Step Editing

Sub steps can be set and removed in voice edit mode:

1. Press and hold the shift button.
2. Use the 16 sequencer buttons to select the main step whose sub steps you want to edit.
3. A flashing LED indicates the selected step.
4. The sub step pattern is shown on the 8 select button LEDs.
5. Set and remove sub steps using the 8 select buttons.

> **Attention!** The step parameter menu always shows the parameters of the last-tweaked step. Setting or removing a sub step causes the display to show that sub step's parameters. For more ergonomic step parameter editing, use Step (Edit) mode.

5.1.4 Sound Editing

- The 7 voice buttons select the active voice (lit LED indicates selection).
- Voice parameters are divided into pages selectable with the 8 select buttons.
- Each voice type has its own parameter set, but the overall page structure is the same.
- Parameters shown in the display are edited with the 4 knobs.

Example — changing the filter frequency of voice 3:

1. Press Voice Button 3 to select the voice.
2. Press Select Button 6 to show the filter page.

3. Use the first knob to adjust the **frq** parameter.

5.2 Performance Mode

This mode is designed for playing songs live. You can chain and change patterns on the fly, trigger manual rolls for each voice, and access the morph and global sample rate effects. Voice editing is not available in this mode.

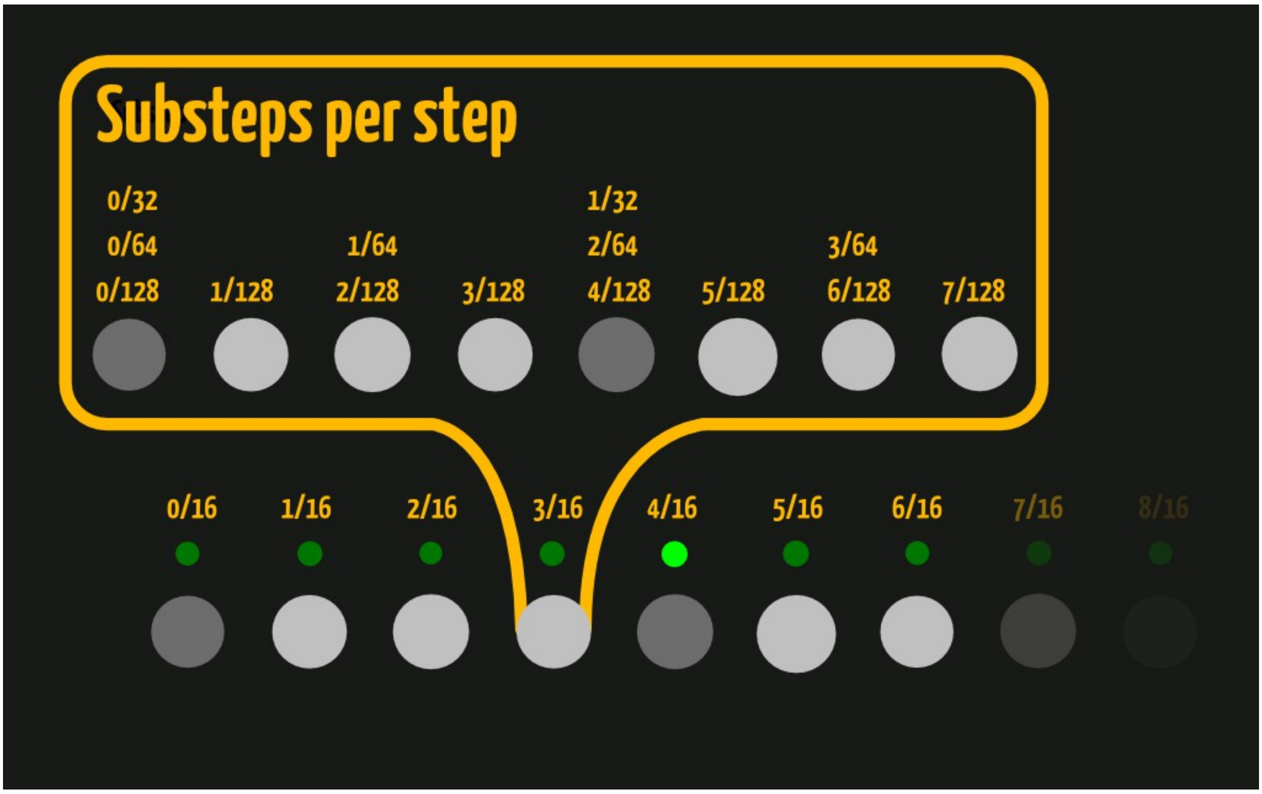

Figure 8. LXR front panel in performance mode.

- **Display (1):** performance parameters.
- **Knobs (2):** change the performance parameters.
- **Sequencer buttons 1–7 (5):** trigger a manual roll for each corresponding voice.
- **Select buttons (7):** change the playing pattern.
- **Voice buttons (9):** mute/unmute voices without holding shift.

5.2.1 Basic Performance Menu

Display	Name	Description
`rol`	Manual roll rate	Trigger rate for the manual roll function of sequencer buttons 1–7.
`mrp`	Morph amount	Ratio between the original sound and the morph target.
`sr`	Global sample rate	A global sample rate reduction effect.
`shu`	Shuffle amount	Sets the amount of global shuffle.

5.2.2 Manual Roll

The first 7 sequencer buttons trigger a manual roll for each corresponding voice. The roll rate is set with the **rol** parameter. Rolls are recorded to the pattern when recording mode is active.

5.2.3 Morph

The morph feature transitions from one preset sound to another. Load any preset from the SD card as a morph target. The **`mrp`** parameter controls the ratio between the original and target sound. As morph increases, the current sound is gradually transformed into the morph target.

For loading a morph target, see the Load and Save Mode section.

5.2.4 Shuffle

The shuffle function shifts the position of every other $\frac{1}{16}$th note.

Original timing: `| X x x x X x x x X x x x X x x x |`

With shuffle applied: `| X xx xX xx xX xx xX xx x |`

5.2.5 Change Pattern

The 8 select buttons change the playing pattern, selecting from the 8 patterns of the loaded pattern set. Pattern changes occur only at the end of a bar (track 1 is the reference track when different track lengths are used).

5.2.6 Chain Patterns

To build longer pattern chains, go to the pattern options screen:

1. Press and hold shift.
2. A flashing LED indicates the selected pattern.
3. The display shows pattern options.

For each pattern you can specify the **next pattern** to play and the **repeat/change measure**.

Change Measure / Repeat (rpt): The change measure is relative to the absolute song position in bars. This ensures pattern chains stay in sync regardless of when you manually switch patterns.

Next Pattern (nxt): Values p1–p8 select a specific pattern. Values r2–r8 select a random pattern between pattern 1 and the selected value (e.g. r4 randomly selects from patterns 1–4).

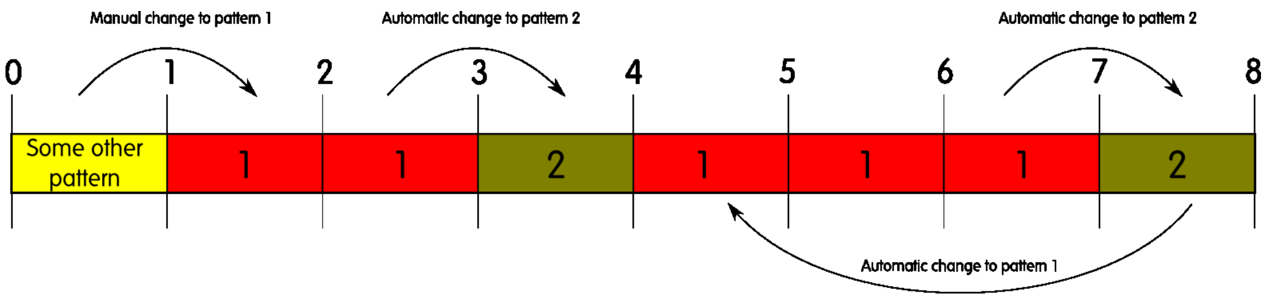

Figure 9. Pattern chain example: pattern 1 plays 3 times, then pattern 2 plays once.

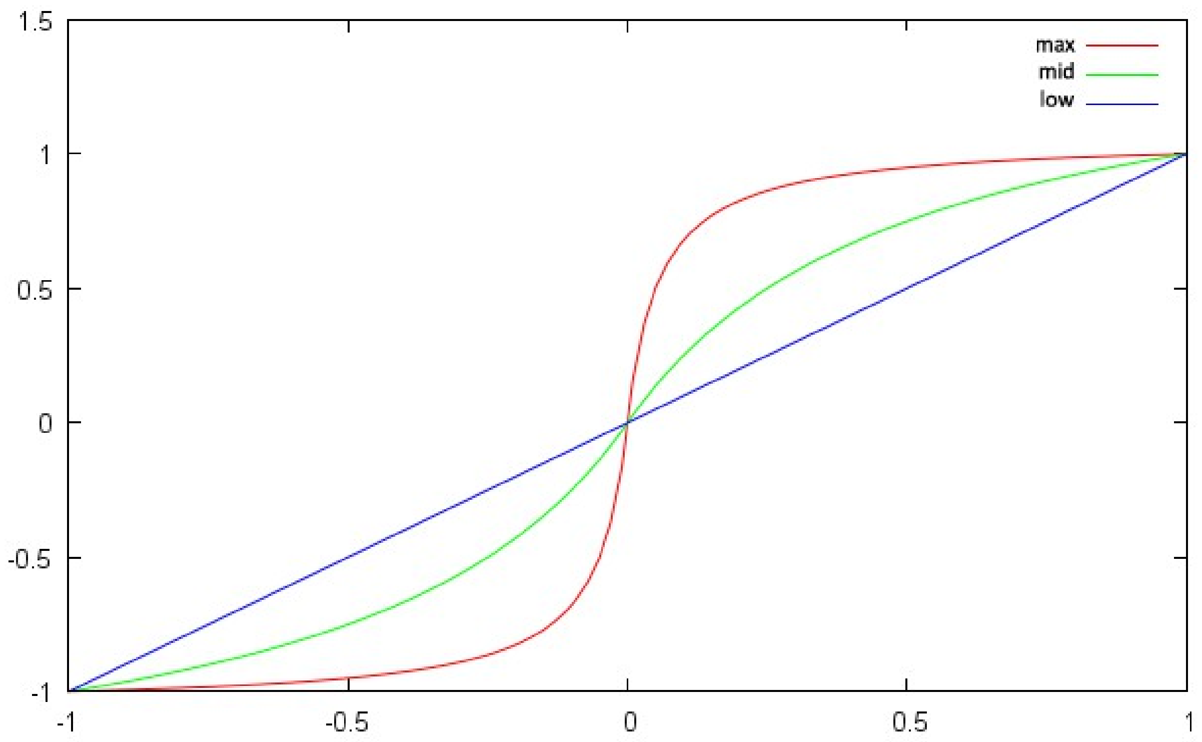

Figure 10. Pattern chain with manual switch: the break always lands on bar 4.

> **Did you know?** When editing multiple chained patterns, set the pattern follow (**`flw`**) parameter in the settings menu to 'Off'. This lets you edit a pattern without it jumping to follow playback.

5.2.7 Follow Mode and Viewed Pattern

The follow mode setting is in the global settings menu (on by default).

When follow mode is active, the viewed pattern always matches the playing pattern — all LEDs and display values update automatically on every pattern change.

When follow mode is off, the user must manually select the viewed pattern. This allows editing a different pattern from the one currently playing.

Selecting the viewed pattern (performance mode only):

1. The currently viewed pattern blinks; the playing pattern shows a solid LED.
2. Hold shift.
3. Use the 8 select buttons to pick the pattern to view/edit.
4. Release shift.

5.3 Step (Edit) Mode

Step edit mode is for editing step parameters more efficiently. Think of it as an inverted voice edit mode with respect to the sequencer.

The display always shows the step parameter page. The sequencer and select buttons are only used to select steps — you can edit step parameters without holding shift.

The 16 sequencer buttons select a main step and show its 8 sub steps on the select button LEDs. The selected step is indicated by a blinking LED.

5.3.1 Setting and Removing Steps

To set or remove steps from the pattern, hold the shift button while pressing the sequencer buttons.

5.4 Load and Save Mode

The 4th mode button opens the load/save page. Pressing it again toggles between load and save.

A push on the Load/Save button brings up the load page:

```
Load:  Sound
[   1] Name
```

Pressing the button again shows the save page:

```
Save:  Sound
[   1] Name                    ok
```

5.4.1 Display Description

- **Row 1:** active mode (Load or Save) and the selected data type (Sound, Pattern, Morph Sound, Settings).
- **Row 2:** active preset number and the name of the selected preset.
- Pages with an `ok` button only execute when you navigate to `ok` and push the encoder.
- Pages without an `ok` button execute as soon as a change occurs.

5.4.2 Menu Navigation

Use the encoder to navigate and change entries. Square brackets `[ ]` show the field that will be changed by the encoder (edit mode). Pushing the encoder switches to selection mode (shows `>`).

> **Did you know?** For the data type field (Sound, Pattern, Morph Sound, Settings), use the first knob to select directly without clicking through encoder modes.

5.4.3 Sound Presets

Sound presets contain all synthesis parameters for the 6 voices.

Loading: sounds load automatically when the preset number changes.

Saving:

1. Select the preset number.
2. Navigate to the name letters.
3. Push the encoder and turn it to change each letter.
4. Navigate to `ok` and push the encoder to save.

Quick naming: while the cursor is in the name area, knob 1 moves the cursor, knob 2 toggles case, and knob 3 scrolls through ASCII characters.

5.4.4 Morph Presets

A morph preset is a normal preset loaded as a morph target.

Loading:

1. Select `MorphSnd` as the data type.
2. Change the preset number — the morph sound loads automatically.
3. Note: no audible change occurs if the **`mrp`** (morph amount) parameter is set to zero.

Saving morph results: select `MorphSnd` on the save screen to save the current blended sound to a new preset slot.

5.4.5 Pattern Sets

Pattern sets are collections of 8 patterns stored in one file.

Loading: navigate to `OK` on the display and push the encoder to start loading (required because loading takes 4–5 seconds). The pattern loads in the background — **the loaded pattern does not play until you press one of the pattern change buttons in performance mode.**

Saving: same procedure as saving a sound preset.

5.4.6 Settings

The settings file contains all parameters from the settings menu. It is automatically loaded at boot. To save current settings, go to save mode, select `Globals`, navigate to `Ok`, and push the encoder.

5.5 Pattern Generator Mode

The pattern generator creates polyrhythms automatically using the Euclidean algorithm.

5.5.1 Accessing the Pattern Generator

1. Press and hold shift.
2. Press Mode Button 2 (Perf/Pattern).
3. A flashing mode 2 LED indicates the pattern generator is active.

5.5.2 Pattern Generator Menu

Display	Name	Description
`len`	Track Length	Length of the track in $\frac{1}{16}$ notes. Each track can have a different length.
`stp`	Number of Steps	The number of active steps to distribute in the pattern.

5.5.3 Generating Patterns

- The generator modifies a single track at a time. Select the track with the 7 voice buttons.
- Changing either **`len`** or **`stp`** immediately generates a new pattern on the active track.

> **Attention!** Existing pattern data is overwritten and there is no undo.

5.6 Global Settings Menu

5.6.1 Accessing the Settings Menu

1. Press and hold shift.
2. Press Mode Button 4 (Load/Save).
3. A flashing mode 4 LED indicates the settings menu is active.

5.6.2 Menu Options

Display	Name	Description
`bpm`	Tempo in BPM	Internal clock tempo. Set to 0 to sync to incoming MIDI clock.
`fet`	Parameter fetch	When active, you must 'fetch' the displayed value with the knob before it changes — prevents value jumps on page switches.
`flw`	Pattern follow	When active, the viewed pattern always matches the playing pattern.
`qnt`	Sequencer Quantisation	Sets the quantisation grid for recorded data.
`ss`	Screensaver	Turns the screensaver on or off.

5.6.3 Follow Mode

When follow mode is active, all viewed pattern data (step LEDs, display values) updates automatically on every pattern change. When off, the user selects the viewed pattern manually — allowing editing of a pattern other than the one playing. The chaselight is only visible when the playing pattern is viewed.

5.6.4 Screensaver

The display is an OLED. To extend its lifetime and prevent burn-in, a screensaver activates after a few minutes of inactivity. Any control touch dismisses it. The screensaver can be disabled in the settings menu.

5.6.5 Loading and Saving Global Settings

Settings are loaded automatically on power-up. To save current settings, go to the save menu, select `Globals`, navigate to `Ok`, and push the encoder.

6 Synth Modules

A closer look at the synthesis modules used across the different voices.

6.1 Oscillator

The oscillators provide 6 different waveforms:

- Sine
- Triangle
- Saw
- Rectangle
- Noise
- Crash

The first 5 are classical analogue waveforms realised using bandlimited wavetable oscillators. The 6th is a TR-909-style crash sample — included as a bonus, since a realistic crash is nearly impossible to synthesise using the available voice structure.

6.2 Filter

The filter is a 2-pole (12 dB) state variable filter (SVF) realised as a non-linear zero-delay-feedback model. It shapes the harmonic content of the sound — for example, a hihat needs a highpass filter to remove low frequencies, while a bandpass is useful for a clap.

There are 3 parameters that control the filter response: frequency, resonance, and drive.

6.2.1 Frequency

The frequency determines the operating point of the filter in the spectrum. A lowpass filter cuts all frequencies above the set cutoff frequency.

6.2.2 Resonance

The resonance controls the feedback path of the filter. Higher resonance settings amplify the frequencies around the operating point (i.e. the cutoff frequency) more and more.

6.2.3 Drive

Controls how hard the filter is driven. More drive yields more distortion. At drive 0 the filter is clean and nearly linear. With higher levels, a soft clipper and the slew-rate limit of the integrators engage. Low settings only affect resonance peaks; high settings distort the whole signal. Because the soft clipper scales down excessive peaks, audible resonance is reduced at high drive settings.

6.2.4 Filter Types

There are several filter types, each with its own characteristics. The plots below show each filter with 3 resonance settings (low = red, medium = green, high = blue). The cutoff frequency is identical in every plot, marked by a vertical dashed line.

Lowpass — Removes high frequencies. All frequencies above the cutoff are reduced gradually.

Figure 11. Lowpass filter response.

Highpass — Removes low frequencies. All frequencies below the cutoff are reduced gradually.

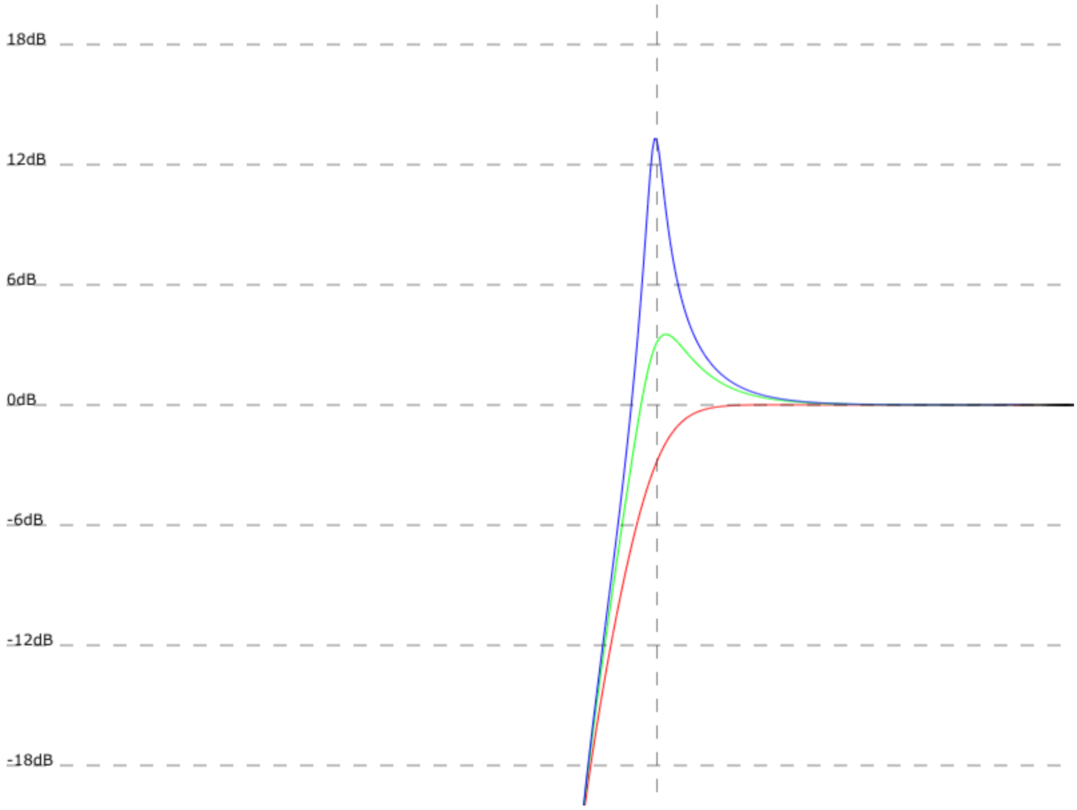

Figure 12. Highpass filter response.

Bandpass — Removes frequencies both above and below the cutoff.

Figure 13. Bandpass filter response.

Unit Gain Bandpass — A scaled bandpass where gain is always adjusted to a maximum of 0 dB. Unlike the normal bandpass, resonance controls the width of the passband rather than the peak amplitude.

Figure 14. Unit gain bandpass filter response.

Notch — Removes frequencies around the cutoff. Resonance controls the width of the stopband.

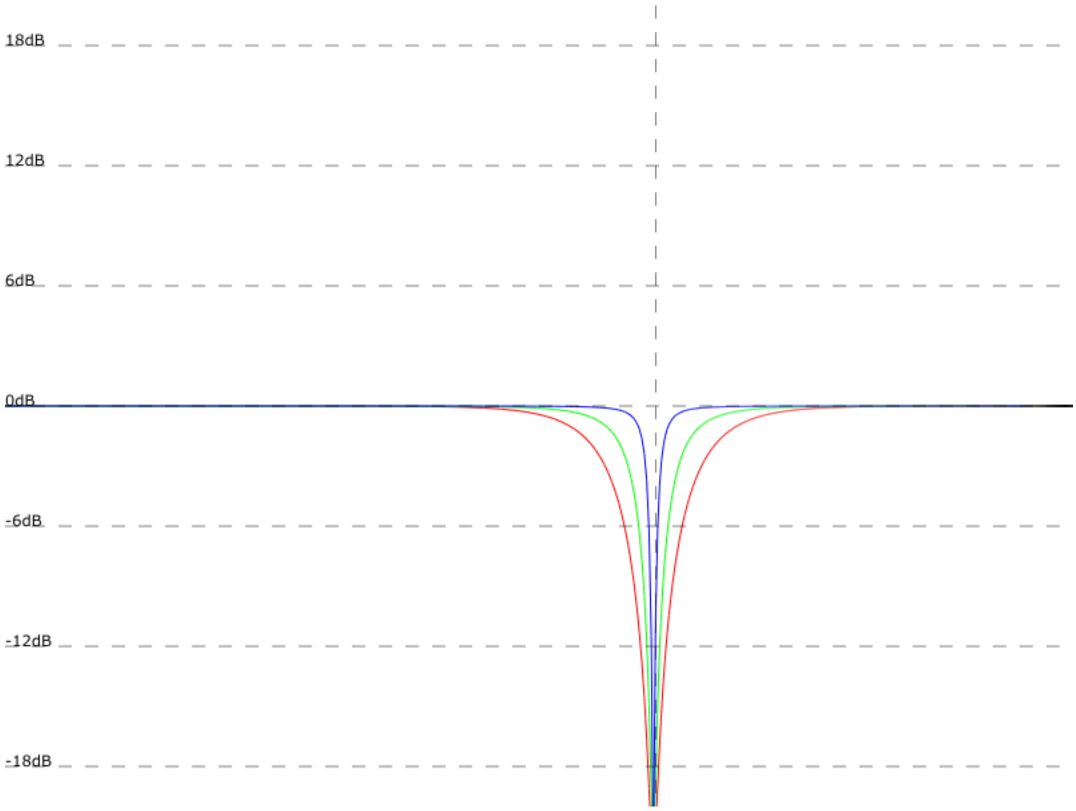

Figure 15. Notch filter response.

Peak — Amplifies frequencies around the cutoff; other frequencies pass nearly unaltered. Resonance controls the amount of amplification.

Figure 16. Peak filter response.

6.3 Envelopes

Envelopes generate a varying control signal to modulate other synthesis parameters. Whenever a voice is triggered, the envelope restarts. The signal rises at the speed set by attack, reaches maximum amplitude, then falls back to zero at the speed set by decay.

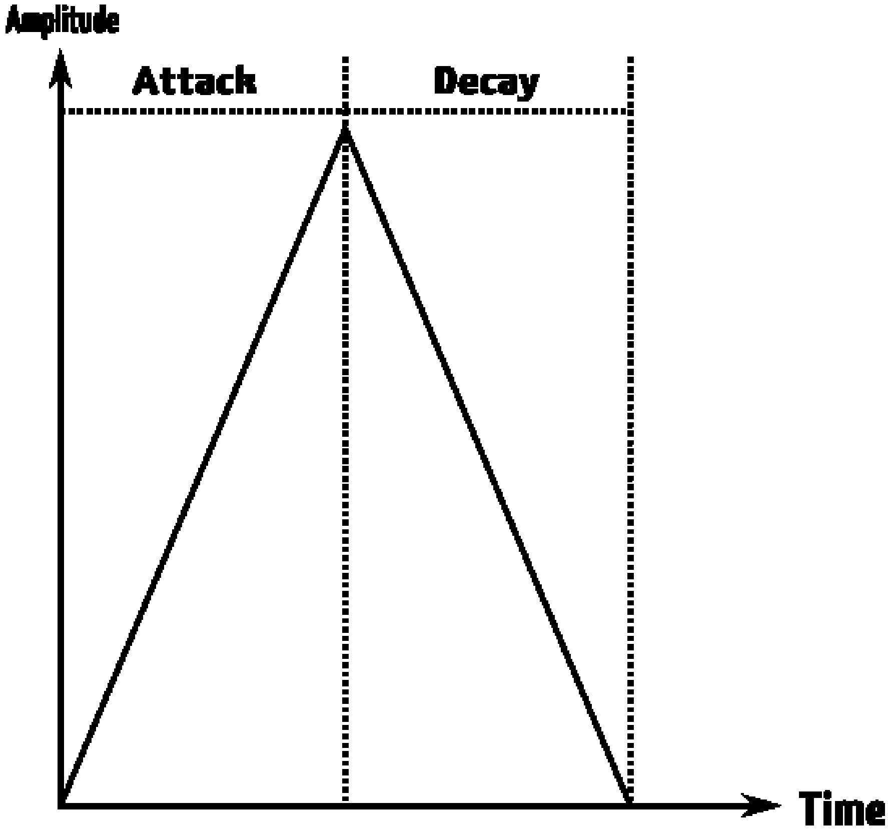

Figure 17. Linear envelope curve (slope = 63).

6.3.1 Attack / Decay Times

These 2 parameters control the duration of each stage. Higher values = longer times. If attack is set to zero, the envelope starts directly in the decay stage.

6.3.2 Slope

The slope parameter controls the curve shape. A value of 63 gives a linear envelope. Lower values produce an exponential curve; higher values produce a logarithmic curve.

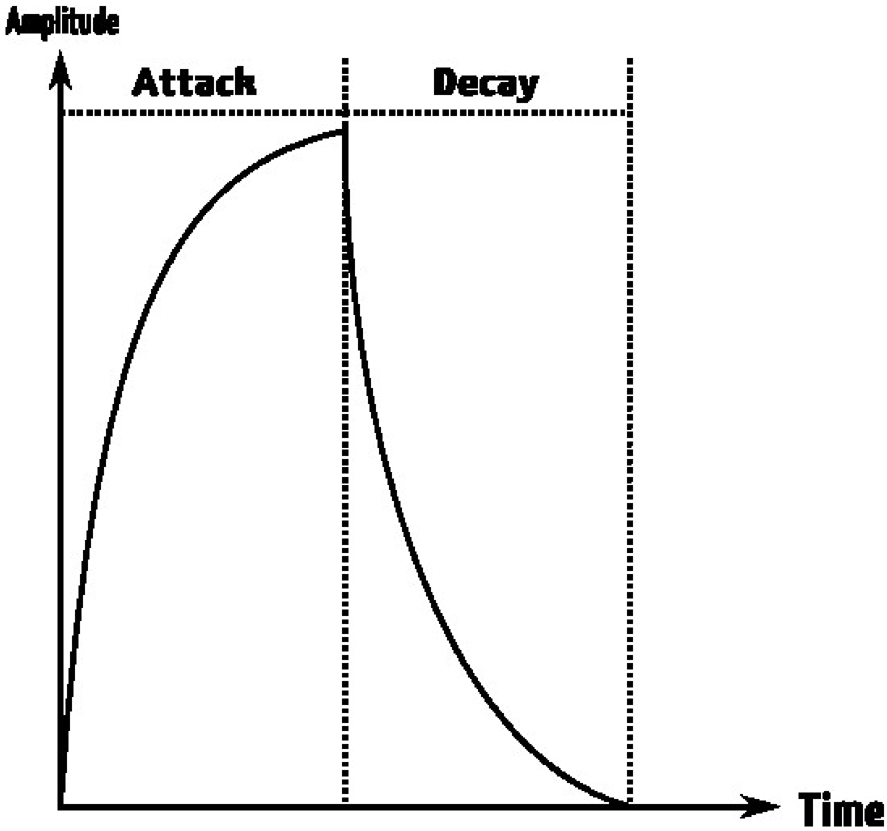

Figure 18. Low slope setting — exponential decay stage.

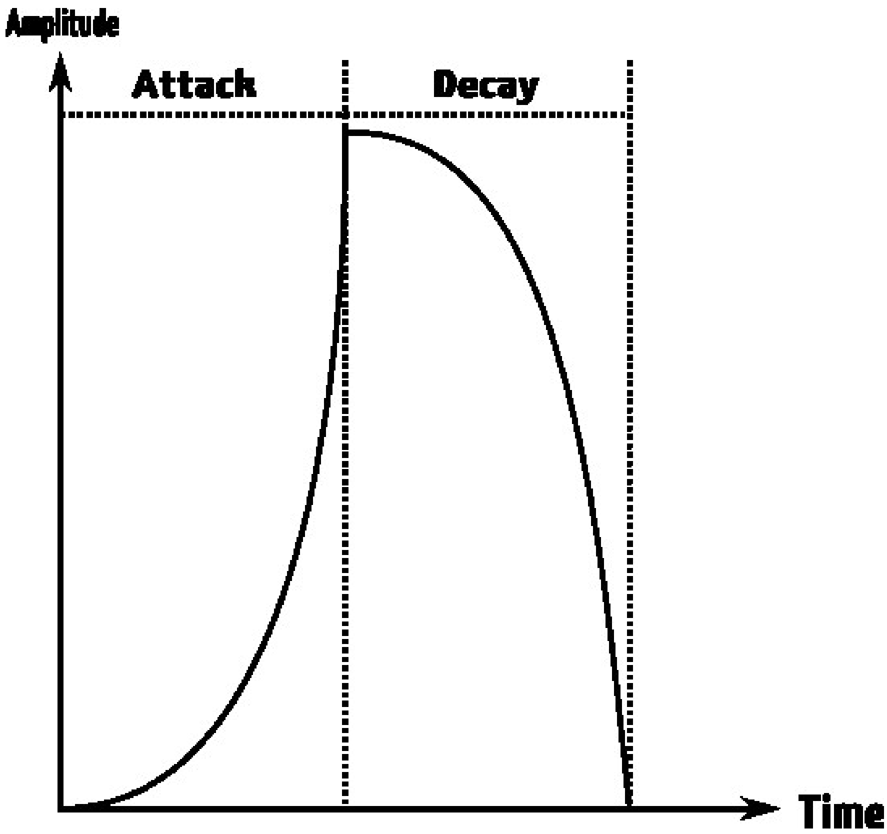

Figure 19. High slope setting — logarithmic decay stage.

6.3.3 Repeat

The repeat feature simulates clap and other sounds with a fuzzy or rattling attack. When the repeat count is set above zero, the slow attack stage is replaced by a repeat stage in which the envelope retriggered rapidly. The attack parameter controls the duration of the complete repeat phase.

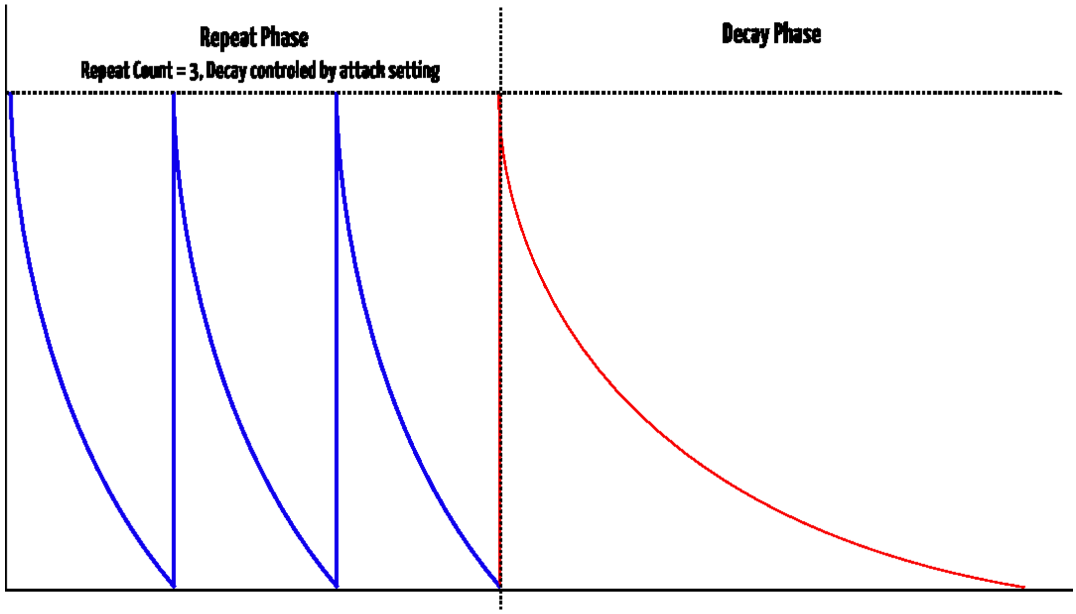

Figure 20. Envelope in repeat mode (count = 3). Blue = repeat phase, red = decay phase.

6.4 Transient Generator

The transient generator shapes the very beginning of a sound. Because the first few milliseconds heavily influence how a sound is perceived, this is a powerful tool for varying the attack character.

6.4.1 Snappy Mode (`Snp`)

An additional fast pitch envelope modulates the oscillator pitch, producing a snap or click. The **`vol`** parameter controls modulation depth; **`frq`** controls the decay time.

6.4.2 Offset Mode (`Off`)

Adjusts the oscillator start phase. At volume 0, oscillators reset to a zero crossing on each trigger. At maximum volume, the waveform starts at its highest amplitude, generating a loud pop.

6.4.3 Sample Mode

A short (~50 ms) 8-bit ROM sample is mixed into the voice output whenever it is triggered. Samples are hardcoded into the firmware and cannot be changed by the user.

Available transient samples:

Display	Name	Description
`Snp`	Snappy Mode	(Pitch envelope — see above.)
`Off`	Offset Mode	(Phase offset — see above.)
`Clk`	Click	Short click transient.
`Ck2`	Click 2	Alternate click.
`Tik`	Tick	A tick-style transient.
`Kik`	Acoustic kick	An acoustic kick drum sample.
`Rim`	Rim shot	A rim shot sample.
`Drp`	Drip	A dripping sound.
`Hat`	Hat	Sampled hi-hat.
`Clp`	Clap	808-style clap.
`Ki2`	Kick 2	Second kick sample.
`Sna`	Snare	A snare sample.
`Tom`	Tom	A tom sample.
`Sp2`	Snap	A finger snap.

6.5 Sample Rate Reduction

The sample rate reducer gradually reduces the sample rate from 44 kHz downward. Reducing the rate gives a lo-fi character. The further it is reduced, the more odd harmonics are added as overtones fold down at the Nyquist frequency — until the sound is completely destroyed.

6.6 Distortion

The distortion unit on the mixer page is a variable waveshaper. It ranges from no distortion through soft saturation up to hard clipping.

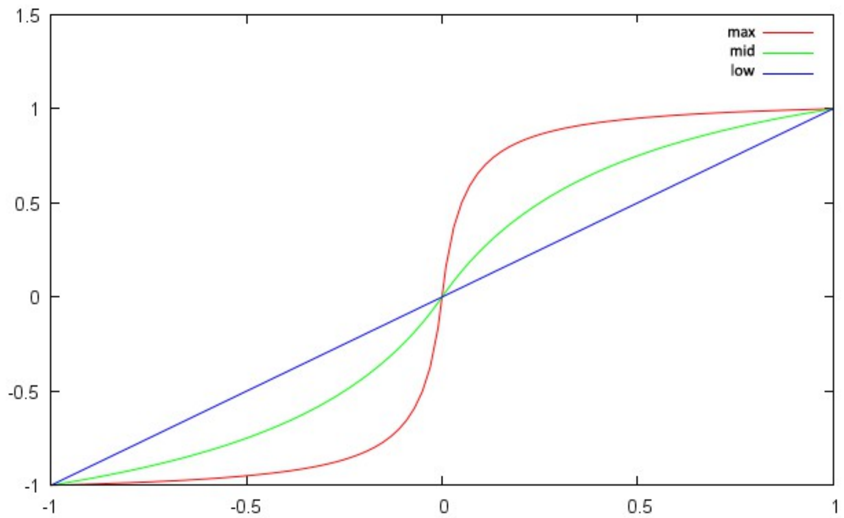

Figure 21. Distortion waveshaper transfer curves: low (blue), mid (green), max (red).

6.7 LFO

Each voice has a low frequency oscillator. LFOs are identical to the audio oscillators but run at a lower maximum rate. They are not audible directly — they modulate other parameter values over time.

Example: use a sine LFO to slowly open and close a voice's filter cutoff.

6.7.1 Setting Up Modulations

The modulation target is selected on the second page of the LFO menu:

```
rtg   off   voi   Dst <
off    0     1      coa
```

The target is specified by 2 parameters: **Voice** and **Destination**. Changing the voice parameter updates the destination list to show all available parameters of the selected voice.

> **Attention!** The LFO modulation scheme differs from most synthesisers. The destination value is *multiplied* by the LFO value, meaning it is always modulated between its current value and zero. It can never exceed the original parameter value. **Modulating a parameter set to zero has no effect.**

The LFO can be synced to the sequencer clock — at $\frac{1}{4}$ note sync, the LFO cycles exactly 4 times per bar. When sync is active, the frequency parameter is ignored.

> **Did you know?** You can use the LFO as an additional envelope by enabling the retrigger feature. Set the retrigger source to the same voice as the modulation destination — the LFO will restart on every played note. Set the LFO frequency so one full cycle is slightly longer than the amplitude envelope, to prevent it retriggering while the note is still sounding.

6.8 Velocity Modulation

Note-on velocity is an important modulation source for bringing life to static patches. Velocity can modulate 2 destinations:

- The voice volume.
- Any single voice parameter.

> **Did you know?** By default, velocity modulates the voice volume. This hard-wired connection can be turned off with the **`vol`** on/off parameter in the modulation menu. Disabling volume modulation frees velocity for use as a third automation track.

6.9 Output Routing

Each voice can be freely routed to any of the 4 output jacks. The jacks operate either as 4 individual mono outputs (pan parameter has no effect) or as 2 stereo pairs.

> **Did you know?** The synth detects whether a cable is inserted in each jack. If a voice is routed to an unconnected output, it automatically plays on the nearest connected output instead.

7 MIDI

The synthesizer has 2 MIDI interfaces: a standard serial interface using DIN connectors, and a MIDI USB interface for use with a computer. Both provide an input/output pair.

7.1 MIDI Channel

The MIDI channel (1–16) is selected in the settings menu. The synth listens to incoming messages on this channel and sends sequencer notes to the MIDI output on the same channel.

7.2 MIDI Sync

The LXR can be synced to external gear using MIDI clock signals. Set the BPM parameter in the settings menu to `0` to enable slave mode. The LXR will then sync its tempo to incoming MIDI clocks, and the Play button has no function — the synth listens for MIDI start/stop signals instead.

When not in MIDI slave mode, the LXR sends its internal clock as MIDI clock messages on both the USB and the DIN MIDI output. This way the LXR can act as a MIDI master and sync other devices. MIDI start/stop messages are sent when the sequencer starts or stops.

7.3 Record Incoming MIDI

You can trigger all voices with external MIDI gear (drum pads, keyboard, etc.). If record mode is active, the played MIDI notes are directly recorded into the corresponding patterns. Note recording can be quantised to a grid.

The default MIDI note numbers for the 7 tracks are:

Voice	MIDI note no.
1	36
2	37
3	38
4	39
5	40
6	41
7	42

7.4 Transmitting Sequencer Data to the MIDI Output

The sequencer currently transmits note-on trigger messages to the MIDI output. The ability to send complete sequencer tracks with pitch information on separate MIDI channels is planned for a future release.

7.5 MIDI CC List

Parameters can be remote-controlled with MIDI CC messages. See the tables below.

7.6 MIDI NRPN List

Since the LXR has more than 127 parameters, some are controlled using NRPN messages.

MIDI CC Table

CC	Function	Voice
0		–
1	Mod wheel	–
2	Waveform Osc 1	1
3	Waveform Osc 1	2
4	Waveform Osc 1	3
5	Waveform Osc 1	4
6	NRPN data entry	–
7	Waveform Osc 1	5
8	Waveform Osc 1	6
9	Coarse tune Osc 1	1
10	Fine tune Osc 1	1
11	Coarse tune Osc 1	2
12	Fine tune Osc 1	2
13	Coarse tune Osc 1	3
14	Fine tune Osc 1	3
15	Coarse tune Osc 1	4
16	Fine tune Osc 1	4
17	Coarse tune Osc 1	5
18	Fine tune Osc 1	5
19	Coarse tune Osc 1	6
20	Fine tune Osc 1	6
21	Waveform mod. OSC	1
22	Waveform mod. OSC	2
23	Waveform mod. OSC	3
24	Waveform mod. Osc 1	5
25	Waveform mod. Osc 2	5
26	Waveform mod. Osc 1	6
27	Waveform mod. Osc 2	6
28	Noise frequency	4
29	Mix osc/noise	4
30	Coarse tune mod. Osc 1	5
31	Coarse tune mod. Osc 2	5
32	Gain mod. Osc 1	5
33	Gain mod. Osc 2	5
34	Coarse tune mod. Osc 1	6
35	Coarse tune mod. Osc 2	6
36	Gain mod. Osc 1	6
37	Gain mod. Osc 2	6
38	Filter frequency	1
39	Filter frequency	2
40	Filter frequency	3
41	Filter frequency	4
42	Filter frequency	5
43	Filter frequency	6
44	Filter resonance	1
45	Filter resonance	2
46	Filter resonance	3
47	Filter resonance	4
48	Filter resonance	5

CC	Function	Voice
49	Filter resonance	6
50	Volume envelope attack	1
51	Volume envelope decay	1
52	Volume envelope attack	2
53	Volume envelope decay	2
54	Volume envelope attack	3
55	Volume envelope decay	3
56	Volume envelope attack	4
57	Volume envelope decay	4
58	Volume envelope attack	5
59	Volume envelope decay	5
60	Volume envelope attack	6
61	Closed hihat decay time	6
62	Open hihat decay time	6
63	Volume envelope slope	1
64	Volume envelope slope	2
65	Volume envelope slope	3
66	Volume EG slope	4
67	Volume EG slope	5
68	Volume EG slope	6
69	Volume EG repeat count	4
70	Volume EG repeat count	5
71	Mod. envelope decay	1
72	Mod. envelope decay	2
73	Mod. envelope decay	3
74	Mod. envelope decay	4
75	Envelope mod. amount	1
76	Envelope mod. amount	2
77	Envelope mod. amount	3
78	Envelope mod. amount	4
79	Mod. envelope slope	1
80	Mod. envelope slope	2
81	Mod. envelope slope	3
82	Mod. envelope slope	4
83	FM amount	1
84	FM frequency	1
85	FM amount	2
86	FM frequency	2
87	FM amount	3
88	FM frequency	3
89	Voice volume	1
90	Voice volume	2
91	Voice volume	3
92	Voice volume	4
93	Voice volume	5
94	Voice volume	6
95	Voice pan	1
96	Voice pan	2
97	Voice pan	3
98	NRPN fine	–

CC	Function	Voice
99	NRPN coarse	–
100	Voice pan	4
101	Voice pan	5
102	Voice pan	6
103	Distortion	1
104	Distortion	2
105	Distortion	3
106	Distortion	4
107	Distortion	5
108	Distortion	6
109	Decimation	1
110	Decimation	2
111	Decimation	3
112	Decimation	4
113	Decimation	5
114	Decimation	6
115	Decimation	all
116	LFO freq	1
117	LFO freq	2
118	LFO freq	3
119	LFO freq	4
120	LFO freq	5
121	LFO freq	6
122	LFO amount	1
123	LFO amount	2
124	LFO amount	3
125	LFO amount	4
126	LFO amount	5
127	LFO amount	6

MIDI NRPN Table

NRPN Nr.	Function	Voice
0	Filter drive	1
1	Filter drive	2
2	Filter drive	3
3	Filter drive	4
4	Filter drive	5
5	Filter drive	6
6	Mix/Mod select	1
7	Mix/Mod select	2
8	Mix/Mod select	3
9	Velocity volume mod. on/off	1
10	Velocity volume mod. on/off	2
11	Velocity volume mod. on/off	3
12	Velocity volume mod. on/off	4
13	Velocity volume mod. on/off	5
14	Velocity volume mod. on/off	6

NRPN Nr.	Function	Voice
15	Velocity mod. amount	1
16	Velocity mod. amount	2
17	Velocity mod. amount	3
18	Velocity mod. amount	4
19	Velocity mod. amount	5
20	Velocity mod. amount	6
21	Velocity mod. destination	1
22	Velocity mod. destination	2
23	Velocity mod. destination	3
24	Velocity mod. destination	4
25	Velocity mod. destination	5
26	Velocity mod. destination	6
27	LFO waveform	1
28	LFO waveform	2
29	LFO waveform	3
30	LFO waveform	4
31	LFO waveform	5
32	LFO waveform	6
33	LFO target voice	1
34	LFO target voice	2
35	LFO target voice	3
36	LFO target voice	4
37	LFO target voice	5
38	LFO target voice	6
39	LFO target	1
40	LFO target	2
41	LFO target	3
42	LFO target	4
43	LFO target	5
44	LFO target	6
45	LFO retrigger	1
46	LFO retrigger	2
47	LFO retrigger	3
48	LFO retrigger	4
49	LFO retrigger	5
50	LFO retrigger	6
51	LFO sync	1
52	LFO sync	2
53	LFO sync	3
54	LFO sync	4
55	LFO sync	5
56	LFO sync	6
57	LFO offset	1
58	LFO offset	2
59	LFO offset	3
60	LFO offset	4
61	LFO offset	5
62	LFO offset	6
63	Filter type	1
64	Filter type	2

NRPN Nr.	Function	Voice
65	Filter type	3
66	Filter type	4
67	Filter type	5
68	Filter type	6
69	Transient generator volume	1
70	Transient generator volume	2
71	Transient generator volume	3
72	Transient generator volume	4
73	Transient generator volume	5
74	Transient generator volume	6
75	Transient generator waveform	1
76	Transient generator waveform	2
77	Transient generator waveform	3
78	Transient generator waveform	4
79	Transient generator waveform	5
80	Transient generator waveform	6
81	Transient generator frequency	1
82	Transient generator frequency	2
83	Transient generator frequency	3
84	Transient generator frequency	4
85	Transient generator frequency	5
86	Transient generator frequency	6
87	Audio output routing	1
88	Audio output routing	2
89	Audio output routing	3
90	Audio output routing	4
91	Audio output routing	5
92	Audio output routing	6

8 Firmware Update

Firmware for both the front AVR and the Cortex-M4 can be updated using the SD card.

8.1 Update Procedure

1. Make sure your SD card is using FAT32. The bootloader can only read FAT32 filesystems.
2. Copy the `FIRMWARE.BIN` file to the root directory of the SD card.
3. Insert the card into the powered-off LXR.
4. Push and hold the encoder, then turn on the synthesizer.

The synth boots in firmware update mode. The display shows:

```
Bootloader v1.1
```

If a firmware image is found on the SD card, the front AVR firmware is updated first:

```
updating...  (1/2)
```

Followed by the mainboard firmware:

```
updating...  (2/2)
```

During the update, the 16 step LEDs show a binary pattern indicating the synth is busy. When the update is complete, the display shows:

```
success!
please reboot...
```

Power the device off and on again. The new firmware version is shown on the normal boot screen.

8.2 Bootloader Error Codes

If problems occur, the bootloader will show one of the following error messages:

Firmware Error — The bootloader could not find the `FIRMWARE.BIN` file. It is either missing from the SD card or the card is not FAT32.

Header error — The `FIRMWARE.BIN` file on the SD card is corrupt or not a valid firmware file. Try downloading the file again.

EOF error — The end of the firmware file was reached before the update finished. The `FIRMWARE.BIN` file is probably corrupted. Try downloading it again.

Mainboard error — The mainboard does not answer. Check that a working mainboard is connected to the front panel.

9 Introduction to Drum Synthesis

This chapter introduces the basics of drum synthesis, focused on the classic analogue sounds as a starting point for further exploration.

9.1 Kicks

The basis of nearly every kick sound is an oscillator combined with a decaying envelope modulating its pitch. Voices 1–3 are ideal for this.

9.1.1 Oscillator

A good starting point is a sine oscillator with a pitch around 30–35.

9.1.2 Amplitude Envelope

A kick starts at maximum amplitude, so set attack to zero. The decay controls the length of the kick — a decay in the range 20–30 with a slope around 25 (exponential) gives a nice average kick. For booming 808-style kicks, increase the decay further.

9.1.3 Pitch Envelope

Set a decay and slope similar to the amplitude envelope as a starting point. All three parameters — decay, slope, and modulation amount — have a large impact on the character of the kick.

9.1.4 Click

The attack transient is an important part of a kick sound. Try different transient generator settings for various clicks and pops, or slightly increase the amplitude attack for softer bass drums.

9.1.5 Tips and Tricks

- You can further shape the attack using the FM oscillator.
- For additional click in the attack, enhance it with a peak filter.

9.2 Snare

The snare is a 2-component sound: the tonal body and the noisy rattle. Voice 4 is designed with both parts in mind — an oscillator for the tonal part and a noise generator with filter for the noise part. Only the noise is routed through the filter, since in most cases you want highpass-filtered noise with an unfiltered drum body.

For most snare sounds the noise should be louder than the tonal part — try setting the oscillator page mix parameter to around 100.

9.2.1 Tonal Part

Similar to a kick but less pronounced. A snare does not need a loud click or deep pitch modulation, and its oscillator frequency should be higher than a kick since it represents a smaller drum.

9.2.2 Noise Part

Highpass-filtered noise with a moderate resonance setting is sufficient for good results.

9.2.3 Amplitude Envelope

Set attack to zero. For the decay:

- Longer decay times with a very exponential slope (as low as 5–10) produce a short hit with a nicely decaying noise tail — the tail acts like a room reverb, giving a more natural result.
- Shorter decay times with a linear or exponential slope give a very dry, direct snare.

9.2.4 Tips and Tricks

- Not every snare needs both tone and noise. Try using only the tonal part with a high pitch modulation envelope to get the classic Kraftwerk electronic zap.

9.3 Clap

A clap sound is noise with a characteristically fuzzy attack — simulating multiple hands clapping with slight timing variations.

9.3.1 Amplitude Envelope

Use the repeat feature to simulate this behaviour. 3–4 repeats works well for a classic clap. The timing differences between claps are very small, so use a short repeat time (attack value below 10). This produces a short burst at the start. For the decay, use an exponential slope with a value below 10.

9.3.2 Oscillator

A single white noise oscillator set to its maximum frequency.

9.3.3 Filter

A bandpass filter with cutoff 60–80 and high resonance gives good results.

9.4 Hihat

9.4.1 Oscillator

Hihats need a complex metallic noise spectrum. Use all 3 oscillators with high frequencies and modulation amounts to build a spectrum you like. A simple white noise oscillator can also work well.

9.4.2 Amplitude Envelope

Set 2 decay times — one for the closed hat, one for the open hat. Both should be short, but the open hihat decay should be longer than the closed. Use a very exponential slope (below 10).

9.4.3 Filter

Use a highpass filter to remove all low frequencies. A high cutoff around 120 works well.

9.5 Cymbal Ride

Cymbals are similar to hihats — the difference is in the envelope and filter settings.

9.5.1 Oscillator

Oscillator settings can be similar to those of the hihat.

9.5.2 Amplitude Envelope

The only real difference from the hihat is a much longer decay time. Use a strong exponential slope. Also consider lowering the voice volume, as cymbal sounds can be very loud.

9.5.3 Filter

A bandpass filter with high resonance and a cutoff above 110 gives good results.

9.5.4 LFO

Cymbals benefit greatly from LFO modulation on the filter frequency. Use a low modulation frequency and a sine wave with a gentle modulation depth — this ensures successive strikes do not sound identical and the frequency spectrum shifts slightly over time.

9.6 Bells

9.6.1 Realistic Bells

Realistic bells are best achieved with FM oscillators. A complex metallic spectrum combined with an exponential decay amplitude envelope and a bandpass filter produces different bell types. The sample rate reducer is very useful for adding a more metallic character.

9.6.2 808-Style Cowbell

The 808 cowbell uses 2 detuned rectangle oscillators through a bandpass filter.

Recommended voice: Drum voice 1–3

FM Page:

- Set mode to Mix (main and FM oscillators are mixed).
- Set amount to 63 (50/50 mix ratio).
- Select rectangle waveform.
- Set frequency to approximately 78.

OSC Page:

- Select rectangle wave.
- Set frequency to 71.

Amplitude Envelope:

- Attack: 0
- Slope: very exponential (around 2–5).
- Decay: 25–50 depending on slope setting.

Filter:

- Bandpass filter.
- Medium resonance around 70.
- Cutoff frequency around 92.

9.7 More Information About Drum Sound Design

Two highly recommended resources:

- The *Sound on Sound* Synth Secrets series contains excellent articles on different drum sounds: http://www.soundonsound.com/search?url=%2Fsearch&Keyword=%22synth+secrets%22
- The Waldorf Attack synthesizer owners manual contains detailed how-tos on drum synthesis: http://waldorf.electro-music.com/attack/docs/attackdrumsounds.pdf

10 Specifications

10.1 Overview

Instruments	6
Sequencer tracks	7
Buttons	39
LEDs	39
Encoder + switch	1
Knobs	4
Display	1 × 16×2 OLED
Audio jacks	4 × 6.3 mm
MIDI jacks	2 (In/Out)
USB	1 × USB-B
Memory card slot	1

10.2 Sequencer

Steps per track	128 (16 main × 8 sub)
Tracks per pattern	7
Patterns per pattern set	8 (loaded simultaneously into memory)
Per-step data	Probability, note, velocity
Automation slots per step	2 parameter changes
Live editing	All actions possible without stopping the sequencer

10.3 Audio

Sample rate	44 kHz
Bit depth	16
Channels	4 × mono

10.4 Hardware

Audio and Sequencer CPU	ARM Cortex-M4 @ 168 MHz
Front panel CPU	Atmega644 @ 20 MHz
Audio codec	2 × CS4344 stereo codec

10.5 Power

7–9V DC power jack, **centre pin positive, at least 600 mA.**

11 Appendix

11.1 References

1. "The Euclidean Algorithm Generates Traditional Musical Rhythms" by G.T. Toussaint, *Proceedings of BRIDGES: Mathematical Connections in Art, Music, and Science*, Banff, Alberta, Canada, July 31–August 3, 2005, pp. 47–56.

11.2 Sample Credits

The following third-party audio samples are embedded in the firmware:

2. **Crash sample** — based on an edited version of `crashBv2.wav` by Matias Reccius (freesound.org), released under CC0 1.0 Universal Public Domain Dedication. Sabian AAX Studio Crash 16", recorded with an Audio Technica AT3040 condenser microphone. Sample was cut, EQ'd, and heavily compressed by Julian Schmidt to approximate a 909 crash.

3. **Tick sound** — sound by patchen, July 14, 2005. http://www.freesound.org/people/patchen/sounds/4102/ Released under Creative Commons Attribution License.

4. **Acoustic kick** — http://www.freesound.org/people/KEVOY/sounds/82279/ Post-processed with EQ, compressor, and amplitude curves by Julian Schmidt. Released under CC0 1.0.

5. **Rim shot** — TR-909 JGB pack, `rs03.wav`, by altemark. http://www.freesound.org/people/altemark/sounds/26672/ Sampled by Janne G:son Berg from his TR-909. Released under Creative Commons Attribution License.

6. **Drip sound** — `Drip2.wav` by Neotone, July 10, 2009. http://www.freesound.org/people/Neotone/sounds/75344/ Released under CC0 1.0.

7. **Hi-hat sample** — Zildjian A-Custom Hi-Hat Cymbals Pedal Chic, by pjcohen. http://www.freesound.org/people/pjcohen/sounds/45668/ Released under CC0 1.0.

8. **808 clap** — provided by shiftr.